

## ORIGINAL ARTICLE

WILEY

# Antimicrobial prophylaxis and the rate of blood stream infections and *Clostridioides difficile* in pediatric stem cell transplantation: A single-center retrospective study

Trad Alrugaib<sup>1</sup> | Abdulrahman Alsultan<sup>2</sup>  | Enas Elbashir<sup>1</sup> | Bayan Albdah<sup>3</sup> |  
Musaed Alharbi<sup>3,4,5</sup> | Mohammed F. Essa<sup>1,3,4</sup> 

<sup>1</sup>Department of Pediatric Hematology/Oncology, King Abdullah Specialist Children's Hospital, Ministry of National Guard Health Affairs, Riyadh, Saudi Arabia

<sup>2</sup>Department of Pediatrics, King Saud University, Riyadh, Saudi Arabia

<sup>3</sup>King Abdullah International Medical Research Center, National Guard Health Affairs, Riyadh, Saudi Arabia

<sup>4</sup>College of Medicine, King Saud bin Abdulaziz University for Health Sciences, Riyadh, Saudi Arabia

<sup>5</sup>Department of Pediatrics, King Abdullah Specialist Children's Hospital, Ministry of National Guard Health Affairs, Riyadh, Saudi Arabia

## Correspondence

Mohammed F. Essa, Department of Pediatric Hematology/Oncology, King Abdullah Specialist Children's Hospital, Ministry of National Guard Health Affairs, Riyadh, Saudi Arabia.  
Email: [essamo@ngha.med.sa](mailto:essamo@ngha.med.sa)

## Abstract

**Background:** The use of prophylactic antibiotics in the pre-engraftment period to minimize the risk of bacteremia is debatable given concerns of *Clostridioides difficile* (*C. diff*), antibiotics resistance, and disruption of gut microbiota.

**Methods:** We retrospectively reviewed the rate and characteristics of bacteremia and *C. diff* infections within the first 100 days post-HSCT in all pediatric patients who received routine antibacterial prophylaxis during HSCT from 2015 to 2018. *C. diff* infection was defined by the presence of three or more unformed stools in 24 h and positive stool test for *C. diff* or its toxins.

**Results:** One hundred and thirty-five (100 allogeneic and 35 autologous) transplants in 123 patients were eligible for analysis. Median age at transplant was 7.1 (range 0.2–13.7), 67 (55%) were women, and diagnosis was malignant condition in 68 patients. Median time to neutrophil engraftment was 18 days (13–23). Cefepime or piperacillin-tazobactam prophylaxis was used in 105 (78%) and 28 (21%) of patients, respectively. Only five (3%) patients had bacteremia during the pre-engraftment period, and 13 (11%) patients developed bacteremia postengraftment. Septic shock was present in only one patient pre-engraftment and was due to gram-negative bacteria. All patients who developed bacteremia received MAC. Thirteen patients (10%) of patients fulfilled *C. diff* infection definition. There was no mortality related to bacterial infections among our patients.

**Conclusions:** The use of antibiotic prophylaxis was associated with low rate of bacteremia in the pre-engraftment period and a 10% risk of *C. diff* infections. More studies are needed to better evaluate the efficacy of antibiotic prophylaxis in HSCT patients.

## KEYWORDS

antibiotic prophylaxis, *C. diff*, GVHD, HSCT

**Abbreviations:** BSI, blood stream infections; BMF, bone marrow failure; *C. diff*, *Clostridioides difficile*; CDI, *Clostridioides difficile* infection; CMV, cytomegalovirus; GVHD, graft versus host disease; HEPA, high-efficiency particulate air; HSCT, hematopoietic stem cell transplant; IVIG, intravenous immunoglobulin; MAC, myeloablative conditioning; PBSC, peripheral blood stem cells; PJP, pneumocystis jiroveci pneumonia; RIC, reduced intensity conditioning; SCA, sickle-cell anemia; SOS, sinusoidal obstruction syndrome; TRM, transplant-related mortality.

## 1 | INTRODUCTION

Infectious complications are one of the leading causes of mortality and morbidity in patients undergoing HSCT.<sup>1</sup> Due to the use of myeloablative and RIC regimens, majority of patients will be neutropenic prior to engraftment. Prophylactic antimicrobial use against bacterial infections is adopted by some centers to decrease the rate of bacterial infections during this period. It is common to use fluoroquinolones such as levofloxacin in this high-risk population.<sup>1</sup>

Bacteremia develops in about 11%–43.4% of pediatric HSCT population.<sup>1,2</sup> This is a wide range because of variable types of HSCT, different diseases, degree of allogeneic matching, the intensity of conditioning regimens, and different centers approach. Patients undergoing haploidentical HSCT or cord blood HSCT are at a highest risk to develop bacteraemia.<sup>3</sup>

The use of prophylactic antibiotic may increase the risk of *Clostridioides difficile* (*C. diff*) infections. According to multiple studies carried out in both pediatric and adult population, the rate of *C. diff* infection ranges from 7%<sup>1</sup> to 55%.<sup>4</sup> The rate of infection is affected by multiple factors including the degree of immunosuppression, the prolonged use of antibiotics, prior *C. diff* colonization,<sup>5</sup> and gut GVHD. Reports suggest that *C. diff* infection increase the risk of gut GVHD.<sup>6</sup>

In our study, we report our practice of using broad-spectrum antibiotic prophylaxis in the absence of access to levofloxacin and describe the rate and severity of bacterial infections in patients post-HSCT including BSI and CDI characteristics. We postulated that we probably encounter less BSI and high rates of CDI compared with literature.

## 2 | PATIENTS AND METHODS

Primary outcome was to identify the rate of bacteremia in patient on antibacterial prophylaxis and to compare it with published data. Secondary outcome was to identify if being on antibacterial antibiotic will increase the risk of developing *C. diff* infections or GVHD. Bacterial BSI or bacteremia was defined by the presence of positive blood culture and clinical symptoms such as fever. Sepsis was defined by the presence of bacteremia in addition to hypotension or inotropic support. The diagnosis of CDI requires: (1) the presence of diarrhea, defined as three or more unformed stools in 24 h, and (2) positive stool test for toxigenic *C. diff* or its toxins (qualitative PCR-based assay for binary toxin and toxicogenic detection).<sup>7–9</sup>

We reviewed retrospectively all children <14 years of age (national cut-limit for care under department of pediatrics) who underwent HSCT at King Abdullah Specialist Children's Hospital between 2015 and 2018 after approval of the institutional review board. Patients with a primary immunodeficiency disease and patients with active infection on active antibiotics treatment prior to transplant were excluded. Charts were reviewed to obtain the following information: age, gender, diagnosis, type of transplant, donor type, stem cell source, conditioning regimen, myeloablative (MAC) versus reduced intensity (RIC), engraftment, antimicrobial prophylaxis, diarrhea, bacteremia, type of organisms, CDI and presence of acute or chronic (GVHD).<sup>7–9</sup>

Engraftment was defined as absolute neutrophil count more than 500/ $\mu$ l for three consecutive days. TRM was defined as death due to any transplantation-related cause other than disease relapse.

### 2.1 | Supportive care

#### 2.1.1 | Bacterial prophylaxis

All patients undergoing HSCT (autologous and allogeneic transplant) received antibacterial prophylaxis when absolute neutrophil count is below 1000/ $\mu$ l until neutrophil engraftment. Antimicrobial choice was as follows:

- Cefepime 50mg/kg/dose IV q8 hours (maximum dose: 6 g/day).
- If patient is allergic to penicillin, ciprofloxacin oral: 15–20mg/kg/dose Q12h (max dose: 750mg/dose) OR IV: 10 mg/kg/dose Q12h (max dose: 400 mg/dose).

Prior to 2016, the antibiotic of choice was piperacillin/tazobactam and was changed to cefepime based on hospital antibiogram data.

All patients stayed in single rooms with HEPA filters and received their conditioning regimen, GVHD prophylaxis, and other supportive measures via indwelling central venous lines and received nutritional support with nasogastric feeding or total parenteral nutrition in case of inadequate intake. Caspofungin or voriconazole were used as antifungal prophylaxis, cefepime was used for antibacterial prophylaxis as above during the pre-engraftment period, acyclovir prophylaxis was used to prevent herpes and varicella-zoster viral reactivation and preemptive strategy was used for CMV infection prevention. PJP prophylaxis was given before HSCT and after HSCT for at least 6 months. Growth factors and IVIG were used when indicated or IgG level <6 g/L. Blood products were transfused to maintain hemoglobin and platelet level per institutional guidelines.

### 2.2 | Statistical analysis

Data were analyzed using the statistical program SAS (version 9.4). Data are presented as median and range for continuous variables, and frequency with a percentage for categorical variables. Fishers exact test was used for association between categorical variables and Wilcoxon two sample test for continuous variables. All statistical tests were considered significant at  $p < .05$ .

## 3 | RESULTS

### 3.1 | Patient and transplant characteristics

Patients underwent 152 transplants during the study period. Seventeen patients were excluded per enrollment criteria; four of them were on treatment for active infection and did not start on

prophylactic antibiotic during transplant period (Figure 1). Table 1 details the clinical characteristics of 123 patients who underwent 135 HSCTs. 67 (55%) of them were female and 56 (45%) were male. Median age was 7.11 (0.22–13.76) years. 100 (74%) allogeneic HSCTs, while 35 (26%) autologous transplants. The source of the stem cells was bone marrow in 96 (96%). Patients underwent HSCT for different indications. Sixty-eight patients had malignant diseases and 55 had nonmalignant diseases. The most common malignant diseases were acute leukemias (19%) followed by neuroblastoma (17%) and medulloblastoma (11%). SCA, thalassemia, and BMF syndromes were the majority (96%) of nonmalignant diseases. The conditioning regimens were myeloablative in (94%) of recipients. Serotherapy was used in (41%) of patients.

### 3.2 | Bacteremia and associated complications

Cefepime was used in 77% of the patients while 21% of them received piperacillin–tazobactam as a result of a change in our institutional guidelines for antibiotic choice in 2016. Ciprofloxacin was used in two patients (1.4%) because of their allergic history to the previously mentioned antibiotics. Median time to neutrophil engraftment was 18 days (13–23). Bacteremia developed in 18 patients, only five (3%) were during the pre-engraftment period. Fever was the presentation in 14 patients and four developed sepsis with (2.9%). The isolated organisms were gram positive in 10 patients, and gram-negative bacteria in eight patients (Table 2).

All patients who developed bacteremia received MAC. However, age, diagnosis, donor type, source, and serotherapy use did not have statistically significant association. The infection rate was higher in the piperacillin–tazobactam group than in the

cefepime group,  $p = .0404$ . Chronic GVHD was higher in patients who had infections (5)  $p = .0178$  (Table 3).

### 3.3 | *Clostridium difficile* infection

Thirteen patients (10%) were symptomatic for *C. diff* and fulfilled CDI criteria out of 26 patients (19.2%) who tested positive for *C. diff*. (Table 4). Only one patient developed CDI postautologous HSCT. No CDI was seen in 10 transplanted infants and only one of 21 patients less than 2 years of age.

Primary disease, prior CDI, GVHD, and serotherapy did not have statistically significant correlation.

### 3.4 | Overall outcome

Median time to neutrophil engraftment was 18 days (13–23). One hundred and eight (87%) of patients remained alive, whereas 13 (13%) of them died because of relapse (11%) or transplant-related reasons (2%) in the last follow-up range between 3 and 7 years.

## 4 | DISCUSSION

In this retrospective cohort study, we found that the rate of bacteremia in patient receiving antimicrobial prophylaxis comparing to literature was only 3.4% in the pre-engraftment period and 14% if the duration is extended to Day +100 post-HSCT. Sepsis and intensive care unit admission was seen only in 2.9% of the patients. This is considered very low compared with 20% risk as per Akinboyo IC et al when antibiotics prophylaxis was not used in 1294 pediatric HSCT patients, although the majority of HSCTs were from cord blood.<sup>2</sup> Perez et al.<sup>10</sup> studied 111 pediatric haploidentical HSCT patients and showed a higher rate of infections (41.4%) in the first 100 days post-HSCT despite using antibacterial prophylaxis.

Cephalosporin was used in our study in the majority of patients as opposed to fluoroquinolones which was studied in two randomized control trials. Alexander et al showed that levofloxacin prophylaxis compared with placebo in acute leukemia and HSCT lead to a significant reduction in bacteremia rate in the acute leukemia group (21.9% vs. 43.4%,  $p = .001$ ) but not statistically significant among HSCT group (11.0% vs. 17.3%,  $p = .06$ ). In post hoc follow-up analysis of the same trial, HSCT group might have a similar effect similar to leukemia group due to difference in neutropenia days and supportive measures during said period (3.6% vs. 5.9%,  $p = .02$ ).<sup>1</sup> Moxifloxacin was studied in adult HSCT population and lead to a significant reduction in bacteremia rate compared with placebo group (8.8% vs. 28.1%,  $p = .042$ ) with no significant increase in side effects.<sup>11</sup> The variability of infection rates as seen in the above studies depends on multiple factors including type of transplant, degree of allogeneic matching, MAC regimens, supportive measures during neutropenia periods, and presence of GVHD. In our study, bacteremia was significantly

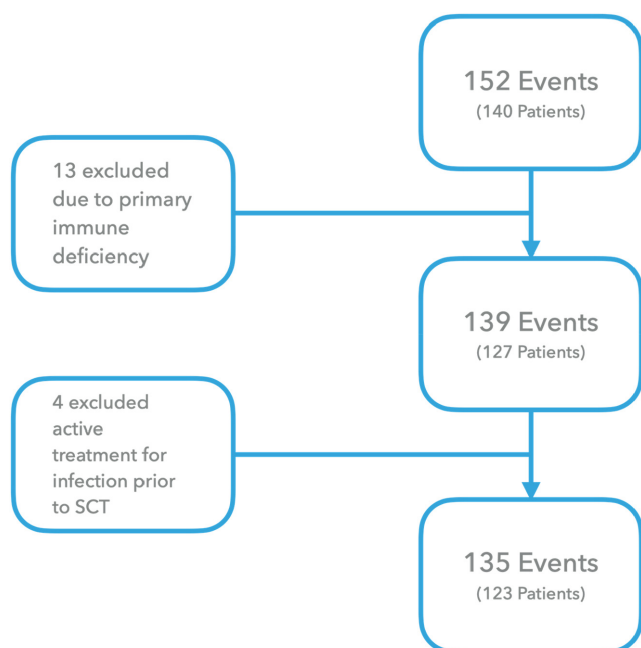


FIGURE 1 Diagram of transplant events included in the study

TABLE 1 Demographics and clinical characteristics

|                               | N    | Range or (%) |
|-------------------------------|------|--------------|
| Median age                    | 7.11 | (0.22–13.76) |
| Gender                        |      |              |
| Male                          | 56   | 45           |
| Female                        | 67   | 55           |
| Diagnosis type                |      |              |
| Malignant                     | 68   | 55           |
| Nonmalignant                  | 55   | 45           |
| Type of transplant            |      |              |
| Allogeneic                    | 100  | 74           |
| Autologous                    | 35   | 26           |
| Donor type of allogeneic HSCT |      |              |
| Matched related               | 75   | 75           |
| Matched unrelated             | 5    | 5            |
| Mismatched/haploidentical     | 17   | 17           |
| Cord                          | 3    | 3            |
| Source of SCT                 |      |              |
| Bone marrow                   | 96   | 96           |
| Peripheral blood              | 1    | 1            |
| Cord blood                    | 3    | 3            |
| Conditioning intensity        |      |              |
| Myeloablative or RT           | 127  | 94           |
| Nonmyeloablative              | 8    | 6            |
| Serotherapy                   | 41   | 41           |

TABLE 2 Clinical characteristics and outcome of patients who developed bacteremia

| Characteristics                             | Patients (%) |
|---------------------------------------------|--------------|
| Bacteremia                                  | 18 (14)      |
| Pre-engraftment                             | 5 (3)        |
| Postengraftment                             | 13 (11)      |
| Gram positive                               | 10 (55)      |
| Coagulase-negative staphylococci 8          | 8            |
| Staphylococcus aureus                       | 1            |
| Enterococcus faecium                        | 1            |
| Gram negative                               | 8 (45)       |
| Enterobacter cloacae                        | 1            |
| Extended Spectrum Beta-Lactamase (ESBL) Sp. | 1            |
| E. coli                                     | 1            |
| Enterobacter aerogenes                      | 1            |
| Klebsiella pneumoniae                       | 1            |
| Pantoea agglomerans                         | 1            |
| Pseudomonas luteola                         | 1            |
| Stenotrophomonas maltophilia                | 1            |
| Presence of septic shock in bacteremia      | 4 (22)       |
| Pre-engraftment/Post-engraftment            | 1/3          |
| Gram positive/gram negative                 | 2/2          |

TABLE 3 Risk factors in patients with bacteremia

| Factor                         | Patients (%) | p value      |
|--------------------------------|--------------|--------------|
| Conditioning intensity .5985   |              |              |
| Myeloablative                  | 18 (100)     |              |
| Reduced intensity              | 0            |              |
| Prophylactic antibiotics .0404 |              |              |
| Cefepime                       | 10/105 (55)  |              |
| Piperacillin/tazobactam        | 8/28 (45)    |              |
| cGVHD                          | 5 (27.7)     | <b>.0178</b> |

Bold value is considered significant if below .050.

TABLE 4 Characteristics of patients with *Clostridium difficile*

| Characteristics                   | Patients (%) |
|-----------------------------------|--------------|
| Positive <i>C. diff</i> infection |              |
| Symptomatic                       | 13 (10)      |
| Asymptomatic                      | 26 (19.2)    |
| Prophylactic antibiotics          |              |
| Cefepime                          | 11/105 (10)  |
| Piperacillin/tazobactam           | 2/28 (7.1)   |

higher in patients receiving piperacillin–tazobactam compared to cefepime ( $p = .0404$ ) as seen in our hospitals antibiogram that lead to the decision of switching to cefepime as the prophylactic antibiotic of choice. It was also noticed that the majority of patients had their events postengraftment while not on antibiotics prophylaxis. However, this still could be explained by the presence of central line devices, Immunosuppression, persistent intestinal microbiota dysbiosis following treatment, GVHD and outpatient care-related factors.

We found that (19.2%) of patients tested positive for *C. diff* but (10%) of patients were symptomatic and met CDI criteria. The rate of CDI in our cohort was comparable or higher than other studies which showed CDI rates range from 2.3% to 8.9%.<sup>1,4,12</sup> This could be due to different types of antimicrobial agents used in different studies. We presume that cefepime and piperacillin/tazobactam are more broad spectrum in affecting gut flora leading to higher CDI. Due to the availability of few patients in the piperacillin/tazobactam group, we could not compare it to cefepime in CDI rates. It was noticed that no CDI was seen in infants and in only one patient for patients less than 2 years of age (15% of study cohort), this could be explained by the pathophysiology of CDI in infants or the primary diseases of those children.

Gender, primary disease, type of transplant, and serotherapy did not have a significant impact on CDI rate in our cohort. Moreover, there was no significant increase in GVHD and CMV reactivation in our patients. Only one mortality was found in the CDI group due to primary disease progression.

GVHD is a well-known HSCT complication that can be influenced or associated with bacteremia and CDI. Infections are risk factors for developing GVHD. It is also described that patients who develop GVHD, will be at higher risk of developing bacteremia due

to loss of anatomical barriers and immunosuppression. We found five patients with positive blood culture whom later developed chronic GVHD ( $p = .0178$ ). Although GVHD can increase the rate of CDI and vice versa we did not find a significant correlation between them in our patients. This is in contrary to Alonso CD et al who described the association between early CDI and the development of gut GVHD in the first-year poststem cell transplant ( $p < .001$ ).<sup>13</sup>

Few limitations were found in our study including the retrospective study design, the small sample size, and the possibility of selection bias as the majority of included patients were transplanted from matched-related donors in one center. Severe dysbiosis in patients receiving broad-spectrum antibiotics is expected, However, no data were collected in regard to the effect on type of flora and dysbiosis in this study population.

In conclusion, antibiotic prophylaxis with cefepime as a single agent in pre-engraftment period was associated with lower rate of bacteremia, sepsis, and acceptable rate of *C. diff* infections. Antibiotic prophylaxis in high-risk allogeneic HSCT patients is warranted. Additional randomized control trial is needed to better evaluate the efficacy and choice of antibiotic prophylaxis. Further studies on antibiotics effects on gut flora and dysbiosis is needed to correlate with patients outcome.

## AUTHORS' CONTRIBUTION

All authors have contributed to the work and have nothing to disclose.

## ACKNOWLEDGMENTS

We would like to thank the team who took care of the patient from Department of Pediatric Hematology/Oncology, Nursing and Microbiology Laboratory Medicine, King Abdulaziz Medical City, Riyadh, Saudi Arabia.

## CONFLICT OF INTEREST

The authors report no conflict of interest.

## DATA AVAILABILITY STATEMENT

Data supplement can be available after communication with corresponding author.

## ORCID

Abdulrahman Alsultan  <https://orcid.org/0000-0001-5577-3228>

Mohammed F. Essa  <https://orcid.org/0000-0003-4860-0796>

## REFERENCES

- Alexander S, Fisher BT, Gaur AH, et al. Effect of levofloxacin prophylaxis on Bacteremia in children with acute Leukemia or undergoing hematopoietic stem cell transplantation a randomized clinical trial. *J Am Med Assoc*. 2018;320(10):995-1004. doi:10.1001/jama.2018.12512
- Akinboyo IC, Young RR, Spees LP, et al. Microbiology and risk factors for hospital-associated bloodstream infections among pediatric hematopoietic stem cell transplant recipients. *Open Forum Infect Dis*. 2020;7(4):1-7. doi:10.1093/ofid/ofaa093
- Atilla E, Atilla PA, Bozdağ SC, Demirel T. A review of infectious complications after haploidentical hematopoietic stem cell transplantations. *Infection*. 2017;45(4):403-411. doi:10.1007/s15010-017-1016-1
- Al-Rawahi GN, Al-Najjar A, McDonald R, et al. Pediatric oncology and stem cell transplant patients with healthcare-associated *Clostridium difficile* infection were already colonized on admission. *Pediatr Blood Cancer*. 2019;66(5):1-5. doi:10.1002/pbc.27604
- Aldrete SDM, Kraft CS, Magee MJ, et al. Risk factors and epidemiology of *Clostridium difficile* infection in hematopoietic stem cell transplant recipients during the peritransplant period. *Transpl Infect Dis*. 2017;19(1):1-6. doi:10.1111/tid.12649
- Callejas-Díaz A, Gea-Banacloche JC. *Clostridium difficile*: deleterious impact on hematopoietic stem cell transplantation. *Curr Hematol Malig Rep*. 2014;9(1):85-90. doi:10.1007/s11899-013-0193-y
- Bagdasarian N, Rao K, Malani PN. Diagnosis and treatment of *Clostridium difficile* in adults: a systematic review. *JAMA*. 2015;313(4):398-408. doi:10.1001/jama.2014.17103
- Debast SB, Bauer MP, Kuijper EJ. European Society of Clinical Microbiology and Infectious Diseases: update of the treatment guidance document for *Clostridium difficile* infection. *Clin Microbiol Infect Off Publ Eur Soc Clin Microbiol Infect Dis*. 2014;20(Suppl 2):1-26. doi:10.1111/1469-0691.12418
- McDonald LC, Coignard B, Dubberke E, Song X, Horan T, Kutty PK. Recommendations for surveillance of *Clostridium difficile*-associated disease. *Infect Control Hosp Epidemiol*. 2007;28(2):140-145. doi:10.1086/511798
- Perez P, Patiño J, Estacio M, Pino J, Manzi E, Medina D. Bacteremia in pediatric patients with hematopoietic stem cell transplantation. *Hematol Transfus Cell Ther*. 2020;42(1):5-11. doi:10.1016/j.htct.2019.05.006
- Vehreschild JJ, Moritz G, Vehreschild MJGT, et al. Efficacy and safety of moxifloxacin as antibacterial prophylaxis for patients receiving autologous haematopoietic stem cell transplantation: a randomised trial. *Int J Antimicrob Agents*. 2012;39(2):130-134. doi:10.1016/j.ijantimicag.2011.10.009
- Salamonowicz M, Ociepa T, Frączkiewicz J, et al. Incidence, course, and outcome of *Clostridium difficile* infection in children with hematological malignancies or undergoing hematopoietic stem cell transplantation. *Eur J Clin Microbiol Infect Dis*. 2018;37(9):1805-1812. doi:10.1007/s10096-018-3316-5
- Alonso CD, Treadway SB, Hanna DB, et al. Epidemiology and outcomes of *Clostridium difficile* infections in hematopoietic stem cell transplant recipients. *Clin Infect Dis*. 2012;54(8):1053-1063. doi:10.1093/cid/cir1035

**How to cite this article:** Alrugaib T, Alsultan A, Elbashir E, Albda B, Alharbi M, Essa MF. Antimicrobial prophylaxis and the rate of blood stream infections and *Clostridioides difficile* in pediatric stem cell transplantation: A single-center retrospective study. *Pediatric Transplantation*. 2023;27:e14375. doi: 10.1111/petr.14375